Oblig 2 DAFE 1000

Finne maksimalpunkt av:

$$f(x) = e^{-\frac{x}{4}} \cdot \tan^{-1} x$$

ved: $f'(x) = 0$

$$(u \cdot v)' = u'v + uv'$$

$$u = e^{-\frac{x}{4}}$$
$$v = \tan^{-1} x$$

Kjerneregel:

$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$

$$u' = -\frac{1}{4} e^{-\frac{x}{4}}$$

$$v' = \arctan(x) = \frac{1}{x^2+1}$$

$$f'(x) = \left(-\frac{1}{4}\tan^{-1}x + \frac{1}{x^2+1}\right)$$

$$\Rightarrow e^{-\frac{x}{4}}\left(-\frac{1}{4}\tan^{-1}x + \frac{1}{x^2+1}\right)$$

Faktorma vore $= 0$

$$f'(x) = 0 \Rightarrow \left(-\frac{1}{4}\tan^{-1}x + \frac{1}{x^2+1}\right) = 0$$

$$\frac{1}{x^2+1} = \frac{1}{4}\tan^{-1}x$$

$$4 \cdot \frac{1}{x^2+1} = 4 \cdot \frac{1}{4}\tan^{-1}x$$

$$\frac{4}{x^2+1} = \tan^{-1}x$$

$$\tan^{-1} x - \frac{4}{x^2+1} = 0$$

Finne x-verdi med numerisk -metode:

$$g(x) = \tan^{-1} x - \frac{4}{x^2+1}$$

$$g(0) = \tan^{-1} \cdot 0 - \frac{4}{0^2+1}$$

$$= 0 - 4 = \underline{\underline{-4}}$$

$$g(4) = \tan^{-1} \cdot 4 - 4 \cdot \frac{4}{x^2+1}$$

$$\approx 1{,}326 - 0{,}235 = \underline{\underline{1{,}09}}$$

x ligger mellom $g(0)$ og $g(4)$

Midtpunkt $= g(2)$

$$g(2) = \tan^{-1} \cdot 2 - 2 \cdot \frac{1}{x^2+1}$$

$$\approx 1{,}107 - 0{,}8 = 0{,}307$$

Ser at toppunkt ligger mellom 0 og 2

Finner $\underline{\underline{x = 1{,}6907}}$ ved bruk av Python.

$$y = f(1{,}6907)$$

$$= e^{-1.6904/4} \cdot \tan^{-1}(1.6907) \approx \underline{\underline{0.6793}}$$